

Terraform workspace

A **Terraform workspace** is a feature in Terraform that allows you to manage **multiple states** (sets of infrastructure) within the same configuration directory.



What is Terraform?

Terraform is an **Infrastructure as Code (IaC)** tool that lets you define and manage your cloud infrastructure using code.



What is a Workspace?

Think of a **workspace** as a separate environment where Terraform keeps its own **state file**. The state file tracks the resources Terraform manages.



Why Use Workspaces?

Imagine you have the same infrastructure setup for:

- **Development**
- **Staging**
- **Production**

Instead of copying your code into three different folders, you can use **workspaces** to manage all three environments from the **same codebase**.



Example

Let's say you have a Terraform configuration for a web server.

You start in the default workspace:

```
terraform workspace show
# Output: default
```

You create a new workspace for development:

```
terraform workspace new dev
```

Now you're in the dev workspace. Terraform will create a **separate state file** for it.

You can switch to another workspace like prod:

```
terraform workspace new prod
```

terraform workspace select prod

Each workspace has its **own state**, so changes in dev won't affect prod.

Terraform Workspace Commands

Command	Description
terraform workspace list	 Lists all the workspaces in your current Terraform project.
terraform workspace show	 Displays the name of the currently active workspace .
terraform workspace new <name>	 Creates a new workspace with the given name and switches to it.
terraform workspace select <name>	 Switches to an existing workspace with the given name.
terraform workspace delete <name>	 Deletes the specified workspace (you cannot delete the current workspace).
terraform workspace	 Shows help information about workspace commands.

Example Usage

terraform workspace list

Output: * default, dev, prod

terraform workspace show

Output: prod

terraform workspace new stage

Creates and switches to 'stage'

terraform workspace select dev

Switches to 'dev'

terraform workspace delete stage

Deletes 'stage' (only if you're not currently in it)